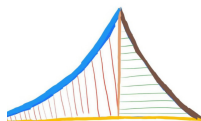


Math for EE

1



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ABOUT THIS BOOK

This book introduces matrices through high school coordinate geometry. This approach makes it easier for beginners to learn Python for scientific computing. All problems in the book are from NCERT mathematics textbooks from Class 9-12. Exercises are from CBSE board exam papers.

The content is sufficient for industry jobs and covers nearly all matrix prerequisites for machine learning. There is no copyright, so readers are free to print and share.

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In this edition, some incorrect solutions were removed. All figures redrawn. More problems added.

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1. GATE AE 2026

A. Formulae

1.1.1

$$\Pr(M) + \Pr(E) - \Pr(ME) + \Pr(M + E)' = 1 \quad (1.1.1.1)$$

1.1.2

$$A + A' = 1, \quad AA' = 0 \quad (1.1.2.1)$$

$$\implies \Pr(A + A') = \Pr(1) = 1 \quad \Pr(AA') = \Pr(0) = 0 \quad (1.1.2.2)$$

1.1.3

$$\Pr(A + A') = \Pr(A) + \Pr(A') \quad (1.1.3.1)$$

1.1.4 Difference rule of probability

$$\Pr(AB') = \Pr(A) - \Pr(AB) \quad (1.1.4.1)$$

Solution: From (1.1.2.1)

$$A = A \cdot 1 = A(B + B') = AB + AB' \quad (1.1.4.2)$$

Also,

$$(AB)(AB') = 0 \quad (1.1.4.3)$$

From (1.1.3.1),

$$\Pr(A) = \Pr(AB + AB') = \Pr(AB) + \Pr(AB') \quad (1.1.4.4)$$

Rearranging, we get:

$$P(AB') = P(A) - P(AB) \quad (1.1.4.5)$$

1.1.5 The Laplace transform is defined as

$$g(t) \xleftrightarrow{\mathcal{L}} G(s) = \int_0^\infty e^{-st} g(t) dt \quad (1.1.5.1)$$

1.1.6

$$g'(t) \xleftrightarrow{\mathcal{L}} sG(s) - g(0) \quad (1.1.6.1)$$

Solution: Applying integration by parts,

$$u = e^{-st}, dv = g'(t) dt \quad (1.1.6.2)$$

$$\implies \int_0^\infty e^{-st} g'(t) dt = [e^{-st} g(t)]_0^\infty - \int_0^\infty (-se^{-st}) g(t) dt \quad (1.1.6.3)$$

$$= -g(0) + s \int_0^\infty e^{-st} g(t) dt \quad (1.1.6.4)$$

yielding (1.1.6.1).

1.1.7

$$g''(t) \xleftrightarrow{\mathcal{L}} s^2 G(s) - sg(0) - g'(0) \quad (1.1.7.1)$$

1.1.8

$$e^{at}g(t) \xleftrightarrow{\mathcal{L}} G(s-a) \quad (1.1.8.1)$$

Solution:

$$\Rightarrow e^{at}g(t) \xleftrightarrow{\mathcal{L}} \int_0^\infty e^{-st} (e^{at}g(t)) dt \quad (1.1.8.2)$$

$$= \int_0^\infty e^{-(s-a)t} g(t) dt \quad (1.1.8.3)$$

yielding (1.1.8.1).

1.1.9 The unit step function is defined as

$$u(t) = \begin{cases} 1 & t > 0 \\ 0 & t < 0 \end{cases} \quad (1.1.9.1)$$

1.1.10

$$u(t) \xleftrightarrow{\mathcal{L}} \frac{1}{s} \quad (1.1.10.1)$$

Solution:

$$\int_0^\infty e^{-st} dt = \left(-\frac{e^{-st}}{s} \right)_0^\infty = \frac{1}{s} \quad \Re(s) > 0 \quad (1.1.10.2)$$

1.1.11

$$tu(t) \xleftrightarrow{\mathcal{L}} \frac{1}{s^2} \quad (1.1.11.1)$$

Solution: Differentiating (1.1.10.2) with respect to s results in (1.1.11.1).

$$\begin{aligned} \cos(at)u(t) &\xleftrightarrow{\mathcal{L}} \frac{s}{s^2 + a^2} \\ \sin(at)u(t) &\xleftrightarrow{\mathcal{L}} \frac{a}{s^2 + a^2} \end{aligned} \quad \Re(s) > a \quad (1.1.11.2)$$

Solution: From (1.1.8.1) and (1.1.10.1),

$$e^{jat}u(t) \xleftrightarrow{\mathcal{L}} \frac{1}{s - ja}, \quad \Re(s) > a \quad (1.1.11.3)$$

$$\Rightarrow \cos(at)u(t) + j \sin(at)u(t) \xleftrightarrow{\mathcal{L}} \frac{s + ja}{(s - ja)(s + ja)} \quad (1.1.11.4)$$

$$= \frac{s}{s^2 + a^2} + j \frac{a}{s^2 + a^2} \quad (1.1.11.5)$$

resulting in (1.1.11.2).

1.1.12 Solve

$$\frac{d^2 y}{dx^2} = -\alpha^2 y, \alpha \neq 0 \quad (1.1.12.1)$$

Solution: Applying the Laplace transform on both sides, and using (1.1.7.1),

$$s^2 Y(s) - sy(0) - y'(0) = -\alpha^2 Y(s) \quad (1.1.12.2)$$

Substituting

$$y(0) = \beta, y'(0) = \gamma \quad (1.1.12.3)$$

in (1.1.12.2),

$$Y(s) = \frac{s\beta + \gamma}{s^2 + \alpha^2} \quad (1.1.12.4)$$

$$= \frac{s\beta}{s^2 + \alpha^2} + \frac{\gamma}{s^2 + \alpha^2} \quad (1.1.12.5)$$

yielding

$$y(t) = \beta \cos(\alpha t) + \frac{\gamma}{\alpha} \sin(\alpha t) \quad (1.1.12.6)$$

from (1.1.11.2).

1.1.13 Euler's Method of Forward Differences. Numerically solve

$$\frac{dy}{dx} = f(x, y), y(x_0) = y_0 \quad (1.1.13.1)$$

Solution: By definition,

$$\left. \frac{dy}{dx} \right|_{x=x_0} = \lim_{h \rightarrow 0} \frac{y(x_0 + h) - y(x_0)}{h} \quad (1.1.13.2)$$

which can be expressed using the notation

$$y'_0 = \frac{y_1 - y_0}{h} \quad (1.1.13.3)$$

Similarly,

$$y''_n = \frac{y'_{n+1} - y'_n}{h} \quad (1.1.13.4)$$

From (1.1.13.3),

$$y'_{n+1} = \frac{y_{n+2} - y_{n+1}}{h}, y'_n = \frac{y_{n+1} - y_n}{h} \quad (1.1.13.5)$$

yielding

$$y''_n = \frac{y_{n+2} - 2y_{n+1} + y_n}{h^2} \quad (1.1.13.6)$$

upon simplification.

1.1.14 Kronecker delta function

$$\delta_n = \begin{cases} 1, & n = 0 \\ 0, & \text{otherwise} \end{cases} \quad (1.1.14.1)$$

1.1.15

$$f_0 \int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} e^{j2\pi n f_0 t} dt = \delta_n \quad (1.1.15.1)$$

1.1.16 The Fourier Series. Let

$$x(t) = \sum_{k=-\infty}^{\infty} x_0(t)(t - nT_0) \quad (1.1.16.1)$$

be a periodic signal with

$$x(t) = \begin{cases} x_0(t), & t \in \left(-\frac{T_0}{2}, \frac{T_0}{2}\right) \\ 0 & \text{otherwise} \end{cases}. \quad (1.1.16.2)$$

Then,

$$x(t) = \sum_{n=-\infty}^{\infty} c_n e^{j2\pi n f_0 t}, \quad f_0 = \frac{1}{T_0} \quad (1.1.16.3)$$

Show that

$$c_n = f_0 \int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} x_0(t) e^{-j2\pi n f_0 t} dt \quad (1.1.16.4)$$

Solution: From (1.1.16.3),

$$\int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} x(t) e^{-j2\pi n f_0 t} dt = \int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} \left(\sum_{k=-\infty}^{\infty} c_k e^{j2\pi k f_0 t} \right) e^{-j2\pi n f_0 t} dt \quad (1.1.16.5)$$

$$\int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} x_0(t) e^{-j2\pi n f_0 t} dt = \sum_{k=-\infty}^{\infty} c_k \int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} e^{j2\pi(k-n)f_0 t} dt \quad (1.1.16.6)$$

$$= \frac{1}{f_0} \sum_{k=-\infty}^{\infty} c_k \delta_{k-n} \quad (1.1.16.7)$$

upon substituting from (1.1.16.2) and (1.1.15.1), yielding (1.1.16.4).

1.1.17 The rectangular function. The unit rectangular function is defined as

$$\text{rect}(t) = \begin{cases} 1 & t \in \left(-\frac{1}{2}, \frac{1}{2}\right) \\ 0 & \text{otherwise} \end{cases} \quad (1.1.17.1)$$

1.1.18 Square Wave. Consider the Fourier series with

$$x_0(t) = \text{rect}(t), \quad T_0 = 2. \quad (1.1.18.1)$$

Show that

$$c_n = \frac{\sin\left(\frac{\pi k}{2}\right)}{\pi k} \quad (1.1.18.2)$$

Solution: Substituting $f_0 = \frac{1}{2}$ in (1.1.16.4),

$$c_n = \frac{1}{2} \int_{-1}^1 e^{-j\pi k t} dt \quad (1.1.18.3)$$

$$= \left[\frac{\sin(\pi k t)}{2\pi k} \right]_{-\frac{1}{2}}^{\frac{1}{2}} \quad (1.1.18.4)$$

The following code verifies the result in Fig. 1.1.18.1. Note that Fig. 1.1.18.1 is continuous while (1.1.18.1) is discontinuous. So (1.1.18.2) provides only an approximation. This is known as the Gibbs' phenomenon.

codes/Fourier.py

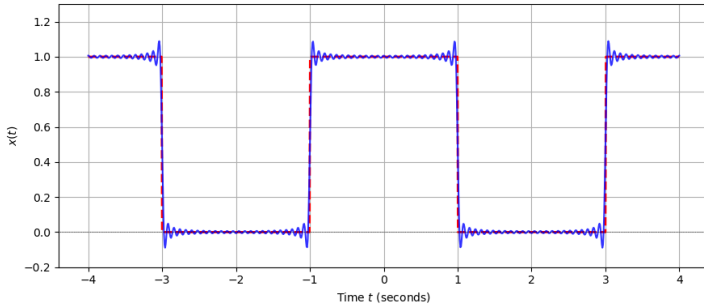


Fig. 1.1.18.1: Gibbs phenomenon

1.1.19 If

$$x(t) \xleftrightarrow{\mathcal{F}} c_n, \quad (1.1.19.1)$$

show that

$$x(t - t_0) \xleftrightarrow{\mathcal{F}} c_n e^{-j2\pi n t_0}, \quad (1.1.19.2)$$

Solution:

$$\int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} x_0(t - t_0) e^{-j2\pi n f_0 t} dt = \int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} x_0(t) e^{-j2\pi n f_0 (t+t_0)} dt \quad (1.1.19.3)$$

$$= e^{j2\pi n f_0 t_0} \int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} x_0(t) e^{-j2\pi n f_0 t} dt \quad (1.1.19.4)$$

after a change of variables. Note that the limits of integration are unchanged due to the periodicity of $x(t)$. Transferring the exponential term in (1.1.19.4) to the L.H.S yields (1.1.19.2).

1.1.20 The standard 2-D basis vectors are defined as

$$\mathbf{e}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \mathbf{e}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \quad (1.1.20.1)$$

Also,

$$\mathbf{1} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad (1.1.20.2)$$

1.1.21 For

$$\mathbf{A} = \mathbf{u}\mathbf{v}^\top, \text{rank}(\mathbf{A}) = 1. \quad (1.1.21.1)$$

where both u and v are n -dimensional real-valued non-zero column vectors.

Solution: Let

$$\mathbf{u} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \mathbf{v} = \begin{pmatrix} 1 \\ 4 \\ 2 \end{pmatrix}. \quad (1.1.21.2)$$

Then

$$\mathbf{A} = \mathbf{u}\mathbf{v}^\top = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \begin{pmatrix} 1 & 4 & 2 \end{pmatrix} \quad (1.1.21.3)$$

$$= \begin{pmatrix} 1 & 4 & 2 \\ 2 & 8 & 4 \\ 3 & 12 & 6 \end{pmatrix} \quad (1.1.21.4)$$

Following row reduction,

$$\mathbf{A} \xrightarrow{R_2 \rightarrow R_2 - 2R_1} \begin{pmatrix} 1 & 4 & 2 \\ 0 & 0 & 0 \\ 3 & 12 & 6 \end{pmatrix} \xrightarrow{R_3 \rightarrow R_3 - 3R_1} \begin{pmatrix} 1 & 4 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \quad (1.1.21.5)$$

yielding (1.1.21.1).

1.1.22

$$|\mathbf{A}^\top| = |\mathbf{A}| \quad (1.1.22.1)$$

1.1.23

$$|\mathbf{AB}| = |\mathbf{A}| |\mathbf{B}| \quad (1.1.23.1)$$

Solution: If $\mathbf{A}$ is invertible

$$\mathbf{A} = \mathbf{E}_1 \mathbf{E}_2 \cdots \mathbf{E}_k \quad (1.1.23.2)$$

where $\mathbf{E}_i$ are invertible matrices. For any single elementary matrix $\mathbf{E}$, it is easy to

verify that

$$|\mathbf{EB}| = |\mathbf{E}| |\mathbf{B}|. \quad (1.1.23.3)$$

$$\therefore |\mathbf{AB}| = |\mathbf{E}_1 \mathbf{E}_2 \cdots \mathbf{E}_k \mathbf{B}| \quad (1.1.23.4)$$

$$= |\mathbf{E}_1| |\mathbf{E}_2 \cdots \mathbf{E}_k \mathbf{B}| \quad (1.1.23.5)$$

$$= |\mathbf{E}_1| |\mathbf{E}_2| \cdots |\mathbf{E}_k| |\mathbf{B}| \quad (1.1.23.6)$$

For $\mathbf{B} = \mathbf{I}$,

$$|\mathbf{A}| = |\mathbf{E}_1| |\mathbf{E}_2| \cdots |\mathbf{E}_k| \quad (1.1.23.7)$$

which upon substituting in (1.1.23.6) yields

$$|\mathbf{AB}| = |\mathbf{A}| |\mathbf{B}| \quad (1.1.23.8)$$

1.1.24

$$|\mathbf{A}^{-1}| = \frac{1}{|\mathbf{A}|} \quad (1.1.24.1)$$

1.1.25 If

$$\mathbf{Ax} = \lambda \mathbf{x}, \quad (1.1.25.1)$$

then λ is an eigenvalue $\mathbf{x}$ of $\mathbf{A}$.

1.1.26 Let $\lambda_1, \lambda_2, \dots, \lambda_i$ be the eigenvalues of a matrix $\mathbf{A}$. Then,

$$\sum_{i=1}^n \lambda_i = \text{Tr}(\mathbf{A}) \quad (1.1.26.1)$$

$$\prod_{i=1}^n \lambda_i = |\mathbf{A}| \quad (1.1.26.2)$$

1.1.27 $\mathbf{A}$ and $\mathbf{A}^\top$ have the same eigenvalues.

Solution: From (1.1.24.1),

$$|\mathbf{A} - \lambda \mathbf{I}| = |(\mathbf{A} - \lambda \mathbf{I})^\top| \quad (1.1.27.1)$$

$$= |\mathbf{A}^\top - \lambda \mathbf{I}|, \quad (1.1.27.2)$$

completing the proof.

1.1.28 Results on the conic sections

a) The equation of a conic with directrix $\mathbf{n}^\top \mathbf{x} = c$, eccentricity e and focus $\mathbf{F}$ is given by

$$\mathbf{g}(\mathbf{x}) = \mathbf{x}^\top \mathbf{V} \mathbf{x} + 2\mathbf{u}^\top \mathbf{x} + f = 0 \quad (1.1.28.1)$$

where

$$\mathbf{V} = \|\mathbf{n}\|^2 \mathbf{I} - e^2 \mathbf{n} \mathbf{n}^\top, \quad (1.1.28.2)$$

$$\mathbf{u} = ce^2 \mathbf{n} - \|\mathbf{n}\|^2 \mathbf{F}, \quad (1.1.28.3)$$

$$f = \|\mathbf{n}\|^2 \|\mathbf{F}\|^2 - c^2 e^2 \quad (1.1.28.4)$$

Solution:

$$\|\mathbf{x} - \mathbf{F}\|^2 = e^2 \frac{(\mathbf{n}^\top \mathbf{x} - c)^2}{\|\mathbf{n}\|^2} \quad (1.1.28.5)$$

$$\implies \|\mathbf{n}\|^2 (\mathbf{x} - \mathbf{F})^\top (\mathbf{x} - \mathbf{F}) = e^2 (\mathbf{n}^\top \mathbf{x} - c)^2 \quad (1.1.28.6)$$

$$\implies \|\mathbf{n}\|^2 (\mathbf{x}^\top \mathbf{x} - 2\mathbf{F}^\top \mathbf{x} + \|\mathbf{F}\|^2) = e^2 (c^2 + (\mathbf{n}^\top \mathbf{x})^2 - 2c\mathbf{n}^\top \mathbf{x}) \quad (1.1.28.7)$$

$$= e^2 (c^2 + (\mathbf{x}^\top \mathbf{n} \mathbf{n}^\top \mathbf{x}) - 2c\mathbf{n}^\top \mathbf{x}) \quad (1.1.28.8)$$

which can be expressed as (1.1.28.1) after simplification.

b) The eccentricity of (1.1.28.1) is given by

$$e = \sqrt{1 - \frac{\lambda_1}{\lambda_2}} \quad (1.1.28.9)$$

Here, λ_1 and λ_2 are the eigenvalues of $\mathbf{V}$

Solution:

From (1.1.28.2), using the fact that $\mathbf{V}$ is symmetric,

$$\mathbf{V}^\top \mathbf{V} = \left(\|\mathbf{n}\|^2 \mathbf{I} - e^2 \mathbf{n} \mathbf{n}^\top \right)^\top \left(\|\mathbf{n}\|^2 \mathbf{I} - e^2 \mathbf{n} \mathbf{n}^\top \right) \quad (1.1.28.10)$$

$$\implies \mathbf{V}^2 = \|\mathbf{n}\|^4 \mathbf{I} + e^4 \mathbf{n} \mathbf{n}^\top \mathbf{n} \mathbf{n}^\top - 2e^2 \|\mathbf{n}\|^2 \mathbf{n} \mathbf{n}^\top \quad (1.1.28.11)$$

$$= \|\mathbf{n}\|^4 \mathbf{I} + e^4 \|\mathbf{n}\|^2 \mathbf{n} \mathbf{n}^\top - 2e^2 \|\mathbf{n}\|^2 \mathbf{n} \mathbf{n}^\top \quad (1.1.28.12)$$

$$= \|\mathbf{n}\|^4 \mathbf{I} + e^2 (e^2 - 2) \|\mathbf{n}\|^2 \mathbf{n} \mathbf{n}^\top \quad (1.1.28.13)$$

$$= \|\mathbf{n}\|^4 \mathbf{I} + (e^2 - 2) \|\mathbf{n}\|^2 \left(\|\mathbf{n}\|^2 \mathbf{I} - \mathbf{V} \right) \quad (1.1.28.14)$$

which can be expressed as

$$\mathbf{V}^2 + (e^2 - 2) \|\mathbf{n}\|^2 \mathbf{V} - (e^2 - 1) \|\mathbf{n}\|^4 \mathbf{I} = 0 \quad (1.1.28.15)$$

Using the Cayley-Hamilton theorem, (1.1.28.15) results in the characteristic equation,

$$\lambda^2 - (2 - e^2) \|\mathbf{n}\|^2 \lambda + (1 - e^2) \|\mathbf{n}\|^4 = 0 \quad (1.1.28.16)$$

which can be expressed as

$$\left(\frac{\lambda}{\|\mathbf{n}\|^2}\right)^2 - (2 - e^2) \left(\frac{\lambda}{\|\mathbf{n}\|^2}\right) + (1 - e^2) = 0 \quad (1.1.28.17)$$

$$\implies \frac{\lambda}{\|\mathbf{n}\|^2} = 1 - e^2, 1 \quad (1.1.28.18)$$

$$\text{or, } \lambda_2 = \|\mathbf{n}\|^2, \lambda_1 = (1 - e^2) \lambda_2 \quad (1.1.28.19)$$

c) (1.1.28.1) represents

- i) a parabola for $|\mathbf{V}| = 0$,
- ii) ellipse for $|\mathbf{V}| > 0$ and
- iii) hyperbola for $|\mathbf{V}| < 0$.

This is a direct result of the following two equations - From (1.1.28.9), we have

$$\frac{\lambda_1}{\lambda_2} = 1 - e^2 \quad (1.1.28.20)$$

From (1.1.26.2), we have

$$|\mathbf{V}| = \lambda_1 \lambda_2 \quad (1.1.28.21)$$

1.1.29 Matrix Determinant Lemma - Consider two column vectors $\mathbf{u}$ and $\mathbf{v}$

$$|\mathbf{I} + \mathbf{u}\mathbf{v}^\top| = 1 + \mathbf{v}^\top \mathbf{u} \quad (1.1.29.1)$$

1.1.30 SHM

- Springs in series: For n springs with spring constants $k_1, k_2, \dots, k_n$ connected in series under an applied tensile force F . - By Hooke's Law, the individual displacements are given by:

$$x_i = \frac{F}{k_i} \forall i \in \{1, 2, \dots, n\} \quad (1.1.30.1)$$

The total displacement in terms of the equivalent spring constant k_{eq} is:

$$x_{\text{total}} = \frac{F}{k_{\text{eq}}} \quad (1.1.30.2)$$

We know that

$$x_{\text{total}} = \sum_{i=1}^n x_i \quad (1.1.30.3)$$

Substituting $x_1, x_2, \dots, x_n$ and x_{total} into (1.1.30.3)

$$\frac{F}{k_{\text{eq}}} = \sum_{i=1}^n \frac{F}{k_i} \quad (1.1.30.4)$$

$$F \left(\frac{1}{k_{\text{eq}}} \right) = F \sum_{i=1}^n \left(\frac{1}{k_i} \right) \quad (1.1.30.5)$$

$$\frac{1}{k_{\text{eq}}} = \sum_{i=1}^n \frac{1}{k_i} \quad (1.1.30.6)$$

- General case of vertical SHM: Consider the following diagram. We need to find out the equation of SHM. When the mass M is at rest, let the system stretch by δ_{st} . Let w be the angular frequency of oscillation.

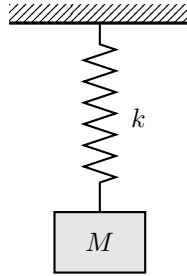


Fig. 1.1.30.1: Spring Mass system in vertical plane

$$\sum F = 0 \implies Mg - k\delta_{\text{st}} = 0 \implies Mg = k\delta_{\text{st}} \quad (1.1.30.7)$$

When the mass is displaced downward by $x(t)$, applying Newton's second law,

$$\sum F = Ma \quad (1.1.30.8)$$

$$\implies M \frac{d^2x}{dt^2} = Mg - k(\delta_{\text{st}} + x) \quad (1.1.30.9)$$

$$\implies M \frac{d^2x}{dt^2} = Mg - k\delta_{\text{st}} - kx \quad (1.1.30.10)$$

The static equilibrium condition is:

$$Mg = k\delta_{\text{st}} \quad (1.1.30.11)$$

Substituting this into (1.1.30.10)

$$M \frac{d^2x}{dt^2} = (k\delta_{\text{st}}) - k\delta_{\text{st}} - kx \quad (1.1.30.12)$$

$$M \frac{d^2 x}{dt^2} = -kx \quad (1.1.30.13)$$

$$\frac{d^2 x}{dt^2} = \frac{-k}{M} x = -w^2 x \quad (1.1.30.14)$$

1.1.31 Results on Gravitation

- Specific angular momentum - Consider a satellite with position vector $\mathbf{r}$ and velocity vector $\mathbf{v}$

$$h_0 = |\mathbf{r} \times \mathbf{v}| \quad (1.1.31.1)$$

- Eccentricity of trajectory

$$e^2 = 1 + \frac{2\varepsilon h_0^2}{\mu^2} \quad (1.1.31.2)$$

- Trajectory of a satellite given its position vector, velocity vector, and gravitational parameter μ : The polar trajectory of a satellite with eccentricity vector $\mathbf{e}$ and semi-latus rectum p_{orb} is given by:

$$r = \frac{p_{\text{orb}}}{1 + e \cos \theta} \quad (1.1.31.3)$$

In vector form:

$$e \cos \theta = \frac{\mathbf{e}^\top \mathbf{x}}{r} \quad (1.1.31.4)$$

Where:

$$\mathbf{x} = \begin{pmatrix} r \cos \theta \\ r \sin \theta \end{pmatrix}. \quad (1.1.31.5)$$

From (1.1.31.3) and (1.1.31.4), we have:

$$r + \mathbf{e}^\top \mathbf{x} = p_{\text{orb}} \implies r = p_{\text{orb}} - \mathbf{e}^\top \mathbf{x} \quad (1.1.31.6)$$

Squaring both sides :

$$r^2 = \mathbf{x}^\top \mathbf{x} = (p_{\text{orb}} - \mathbf{e}^\top \mathbf{x})^2 = p_{\text{orb}}^2 - 2p_{\text{orb}} \mathbf{e}^\top \mathbf{x} + (\mathbf{e}^\top \mathbf{x})^2 \quad (1.1.31.7)$$

Rearranging yields:

$$\mathbf{x}^\top (\mathbf{I} - \mathbf{e} \mathbf{e}^\top) \mathbf{x} + 2(p_{\text{orb}} \mathbf{e})^\top \mathbf{x} - p_{\text{orb}}^2 = 0 \quad (1.1.31.8)$$

Comparing this with (1.1.28.1):

- Matrix $\mathbf{V}$: $\mathbf{V} = \mathbf{I} - \mathbf{e} \mathbf{e}^\top$
- Vector $\mathbf{u}$: $\mathbf{u} = p_{\text{orb}} \mathbf{e}$
- Scalar f : $f = -p_{\text{orb}}^2$

From result (1.1.29.1), we have:

$$|\mathbf{V}| = 1 - \mathbf{e}^\top \mathbf{e} = 1 - e^2 \quad (1.1.31.9)$$

From the sign of $|\mathbf{V}|$, we get the trajectory.

B. Problems

1.2.1 How many 3-digit numbers can be formed using three distinct single digit prime numbers? (AE 2026)

- a) 64 b) 24 c) 12 d) 4

Solution: There are 4 single digit prime numbers: 2,3,5 and 7. The desired numbers are given by

$${}^4P_3 = \frac{4!}{1!} = 24 \quad (1.2.1.1)$$

1.2.2 In a group of students, 10 students like Mathematics, 12 students like English, 4 students like both Mathematics and English, and 6 students like neither Mathematics nor English. The number of students in the group is _____. (AE 2026)

- a) 18 b) 20 c) 24 d) 32

Solution: Let N be the total number of students in the group and

- M : The event that the student likes Mathematics.
- E : The event that the student likes English.

The respective probabilities are expressed as

$$\Pr(M) = \frac{10}{N}, \quad \Pr(E) = \frac{12}{N}, \quad \Pr(ME) = \frac{4}{N}, \quad \Pr((M+E)') = \frac{6}{N} \quad (1.2.2.1)$$

From (1.1.2.2)

$$\Pr(M+E) + \Pr((M+E)') = 1 \quad (1.2.2.2)$$

From (1.1.1.1)

$$\Pr(M) + \Pr(E) - \Pr(ME) + \Pr((M+E)') = 1 \quad (1.2.2.3)$$

Substituting from (1.2.2.1) in the above,

$$\frac{10}{N} + \frac{12}{N} - \frac{4}{N} + \frac{6}{N} = 1 \quad (1.2.2.4)$$

$$\implies N = 24 \quad (1.2.2.5)$$

1.2.3 Given a square of side $s = 4\text{cm}$ and two circles, one with known radius $r_1 = 1\text{cm}$ and another with unknown radius r_2 , such that both of them touch each other externally and also the square, find the value of r_2 . Also describe how the figure in the problem can be constructed. (AE 2026)

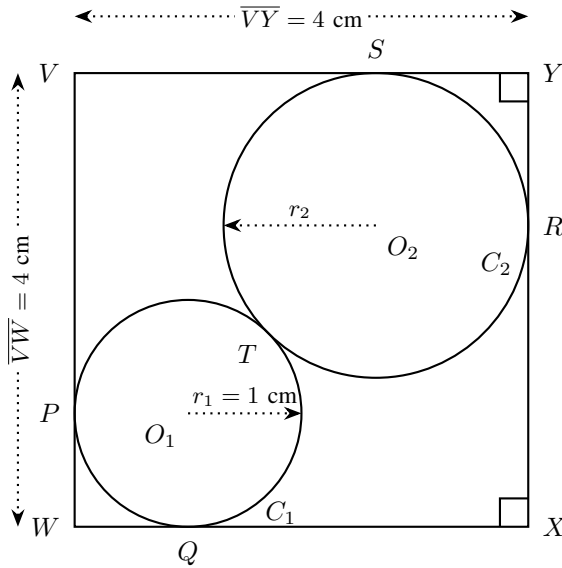


Fig. 1.2.3.1

Solution: In Fig. 1.2.3.1, using (1.1.20.1) and (1.1.20.2),

$$\mathbf{W} = \mathbf{0}, \mathbf{X} = s\mathbf{e}_1, \mathbf{Y} = s\mathbf{1}, \mathbf{V} = s\mathbf{e}_2, \mathbf{O}_1 = r_1\mathbf{1}, \mathbf{O}_2 = (s - r_2)\mathbf{1} \quad (1.2.3.1)$$

Upon substituting from (1.2.3.1)

$$\|\mathbf{O}_2 - \mathbf{O}_1\| = (s - r_1 - r_2)\mathbf{1} = r_1 + r_2 \quad (1.2.3.2)$$

$$\Rightarrow \sqrt{2}(s - r_1 - r_2) = r_1 + r_2 \quad (1.2.3.3)$$

$$\Rightarrow r_2 = \frac{\sqrt{2}s - (\sqrt{2} + 1)r_1}{\sqrt{2} + 1}. \quad (1.2.3.4)$$

Substituting the values of s and r_1 ,

$$r_2 = (7 - 4\sqrt{2}) \text{ cm} \quad (1.2.3.5)$$

The following code generates Fig. 1.2.3.2.

codes/Circle.py

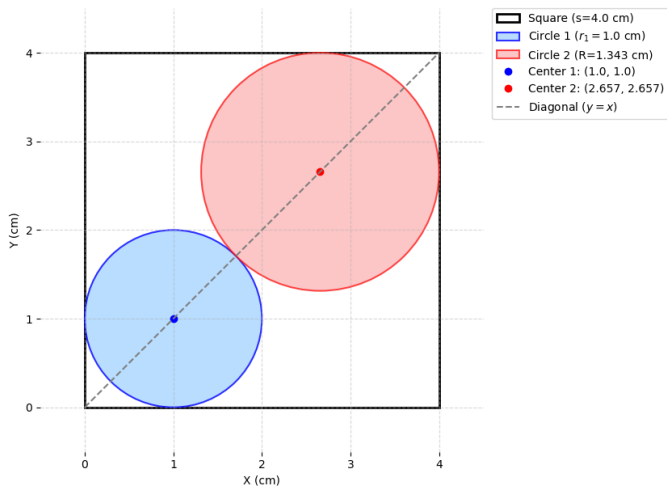


Fig. 1.2.3.2: Square with Tangent Circles

1.2.4 Consider the contour C shown in the figure below. For the vector $\mathbf{F} = (x + 2y) \hat{e}_x + (2x + 4y) \hat{e}_y$, the integral $\oint_C \mathbf{F} \cdot d\mathbf{l} = \underline{\hspace{2cm}}$. Here $d\mathbf{l}$ represents an infinitesimal length along the contour C . (AE 2026)

a) 0

b) 2

c) 4

d) 6

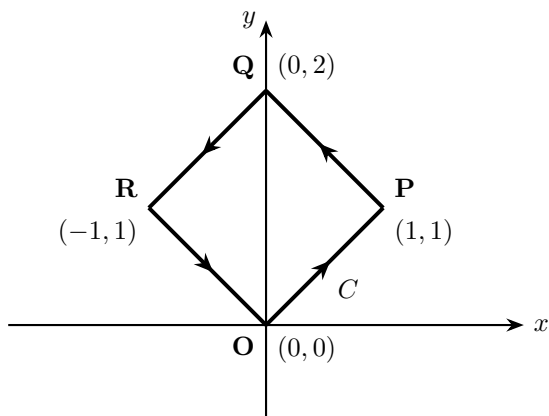


Fig. 1.2.4.1: Contour C

Solution:

$$\therefore \mathbf{F}^\top d\mathbf{l} = (x + 2y)dx + (2x + 4y)dy, \quad (1.2.4.1)$$

$$\int \mathbf{F} \cdot d\mathbf{l} = \int x dx + 2 \int (x dy + y dx) + \int 4y dy \quad (1.2.4.2)$$

$$= \frac{x^2}{2} + 2 \int d(xy) + 2y^2 \quad (1.2.4.3)$$

$$= \frac{x^2}{2} + 2(xy) + 2y^2 \quad (1.2.4.4)$$

$$\therefore \int_{(0,0)}^{(1,1)} \mathbf{F} \cdot d\mathbf{l} = \left[\frac{x^2}{2} \right]_0^1 + 2[xy]_{(0,0)}^{(1,1)} + [2y^2]_0^1 = \frac{9}{2} \quad (1.2.4.5)$$

$$\int_{(1,1)}^{(0,2)} \mathbf{F} \cdot d\mathbf{l} = \left[\frac{x^2}{2} \right]_1^0 + 2[xy]_{(1,1)}^{(0,2)} + [2y^2]_1^2 = \frac{7}{2} \quad (1.2.4.6)$$

$$\int_{(0,2)}^{(-1,1)} \mathbf{F} \cdot d\mathbf{l} = \left[\frac{x^2}{2} \right]_0^{-1} + 2[xy]_{(0,2)}^{(-1,1)} + [2y^2]_2^1 = -\frac{15}{2} \quad (1.2.4.7)$$

$$\int_{(-1,1)}^{(0,0)} \mathbf{F} \cdot d\mathbf{l} = \left[\frac{x^2}{2} \right]_{-1}^0 + 2[xy]_{(-1,1)}^{(0,0)} + [2y^2]_1^0 = -\frac{1}{2} \quad (1.2.4.8)$$

$$\therefore, \oint_C \mathbf{F} \cdot d\mathbf{l} = 0 \quad (1.2.4.9)$$

upon summing up the above integrals. The following code simulates the result. in Fig. 1.2.4.2.

codes/Line_Integral.py

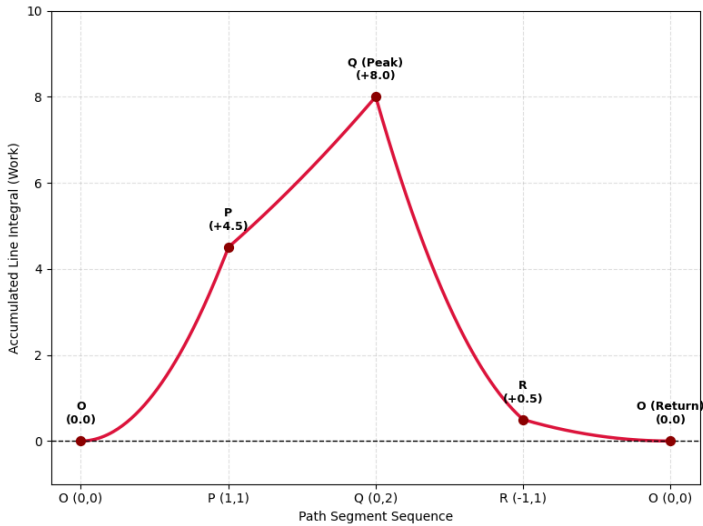


Fig. 1.2.4.2

1.2.5 The Fourier series representation of a square wave is shown in the figure below. The fluctuations seen near $x = \pm 1$ are named after which one of the following scientists? (AE 2026)

- a) Cauchy b) Fourier c) Gibbs d) Laplace

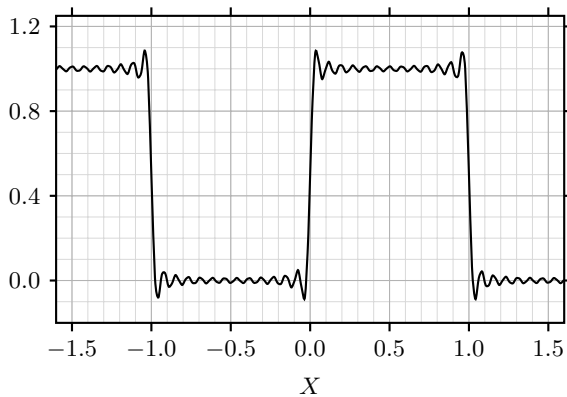


Fig. 1.2.5.1: Gibbs phenomenon

Solution: Fig. 1.2.5.1 can be expressed as a periodic function with parameters

$$x_0(t) = \text{rect} \left(t - \frac{1}{2} \right), T_0 = 2 \quad (1.2.5.1)$$

From (1.1.19.2) and (1.1.18.2), substituting $t_0 = \frac{1}{2}$,

$$c_n = \frac{\sin\left(\frac{\pi k}{2}\right)}{\pi k} e^{-j\pi n} \quad (1.2.5.2)$$

The Gibbs phenomenon is explained in Fig. 1.1.18.1.

1.2.6 If a matrix can be written as $\mathbf{A} = \mathbf{u}\mathbf{v}^\top$, where both u and v are n -dimensional real-valued non-zero column vectors, then the rank of the matrix $\mathbf{A}$ is? (AE 2026)

Solution: From (1.1.21.1), the rank is 1.

1.2.7 An $n \times n$ square matrix $\mathbf{A}$ satisfies $\mathbf{A}^\top = \mathbf{A}^{-1}$. The determinant of this matrix may take which of the following value(s)? (AE 2026)

- a) 1 b) -1 c) n d) 0

Solution: From the given information,

$$|\mathbf{A}^\top| = |\mathbf{A}^{-1}| \quad (1.2.7.1)$$

$$\Rightarrow |\mathbf{A}| = \frac{1}{|\mathbf{A}|} \quad (1.2.7.2)$$

$$\therefore |\mathbf{A}| = \pm 1 \quad (1.2.7.3)$$

substituting (1.1.22.1) and (1.1.24.1) in (1.2.7.1)

1.2.8 Given the velocity and position vectors of a satellite along with its gravitational parameter $\mu = GM$, determine the trajectory of the satellite. (GATE AE 2026)

- a) Circle
b) Hyperbola
c) Parabola
d) Straight Line

Position vector:

$$\mathbf{r} = 8000\hat{p} + 9000\hat{q} \text{ km} \quad (1.2.8.1)$$

Velocity vector:

$$\mathbf{v} = -6\hat{p} + 6\hat{q} \text{ km/s} \quad (1.2.8.2)$$

Gravitational parameter:

$$\mu = 398600 \text{ km}^3/\text{s}^2 \quad (1.2.8.3)$$

Solution: We are given :

$$\mathbf{r} = \begin{pmatrix} 8000 \\ 9000 \end{pmatrix}, \mathbf{v} = \begin{pmatrix} -6 \\ 6 \end{pmatrix} \quad (1.2.8.4)$$

From (1.1.31.1), Specific Angular Momentum h_0

$$= |8000(6) - 9000(-6)| = 102000 \text{ km}^2/\text{s} \quad (1.2.8.5)$$

From (1.1.31.2), eccentricity squared e^2

$$= 1 + \frac{2(2.898)(102000)^2}{(398600)^2} \approx 1 + 0.3795 = 1.3795 \quad (1.2.8.6)$$

From (1.1.31.9), we have:

$$|\mathbf{V}| = 1 - 1.3795 = -0.3795 \quad (1.2.8.7)$$

From 1.1.28(c)iii, we can conclude that the trajectory of the satellite is a Hyperbola. The following code simulates the result.

codes/Satellite.py

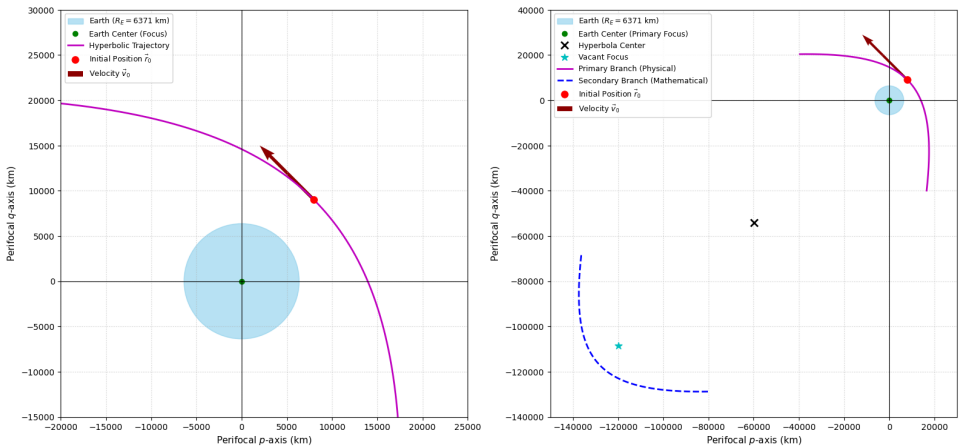


Fig. 1.2.8.1: Trajectory of the Satellite

1.2.9 For the matrix $\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, if the relation $a + b = c + d$ holds and $a, b, c, d \neq 0$, then which one of the following statements about A is FALSE? (AE 2026)

- a) $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ is an eigenvector
 b) $\lambda = a + b$ is an eigenvalue
 c) $\lambda = d - b$ is an eigenvalue
 d) $\lambda = d + b$ is an eigenvalue

Solution: Let

$$a + b = c + d = k \quad (1.2.9.1)$$

Then,

$$\begin{aligned} \begin{pmatrix} a & b \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} &= k \\ \begin{pmatrix} c & d \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} &= k \end{aligned} \quad (1.2.9.2)$$

yielding

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = k \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad (1.2.9.3)$$

$\therefore \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ is an eigenvector and $a + b$ is an eigenvalue from (1.1.25.1) . From (1.2.9.1),

$$a - c = d - b = -(b - d) = l \quad (1.2.9.4)$$

Then,

$$\begin{pmatrix} a & c \\ b & d \end{pmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = l \begin{pmatrix} 1 \\ -1 \end{pmatrix} \quad (1.2.9.5)$$

Thus, $\begin{pmatrix} 1 \\ -1 \end{pmatrix}$ is an eigenvector and $d - b$ is the 2nd eigenvalue.

1.2.10 Consider the differential equation:

$$\frac{d^2 y}{dx^2} + 2 \frac{dy}{dx} + y = 0 \quad (1.2.10.1)$$

with initial conditions $y|_{x=0} = 0$ and $\left. \frac{dy}{dx} \right|_{x=0} = 1$. Find the value of the slope $\frac{dy}{dx}$ at $x = \ln(2)$ rounded off to three decimal places. (AE 2026)

Solution:

a) Theoretical solution - Let

$$y(x) \xleftrightarrow{\mathcal{L}} Y(s) \quad (1.2.10.2)$$

Taking the Laplace transform on both sides of (1.2.10.1)

$$\mathcal{L} \left\{ \frac{d^2 y}{dx^2} \right\} + 2\mathcal{L} \left\{ \frac{dy}{dx} \right\} + \mathcal{L}\{y\} = 0 \quad (1.2.10.3)$$

From equations (1.1.6.1) and (1.1.7.1):

$$\left[s^2 Y(s) - sy(0) - y'(0) \right] + 2[sY(s) - y(0)] + Y(s) = 0 \quad (1.2.10.4)$$

Substituting the given initial conditions:

$$y(0) = 0, y'(0) = 1 \quad (1.2.10.5)$$

$$(s^2 Y(s) - 1) + 2sY(s) + Y(s) = 0 \quad (1.2.10.6)$$

$$(s^2 + 2s + 1)Y(s) = 1 \quad (1.2.10.7)$$

$$Y(s) = \frac{1}{(s+1)^2} \quad (1.2.10.8)$$

Taking the inverse Laplace transform by applying the first frequency shift theorem((1.1.8.1)) on (1.1.11.1) gives the solution $y(x)$:

$$y(x) = xe^{-x} \quad (1.2.10.9)$$

Differentiating $y(x)$ with respect to x :

$$\frac{dy}{dx} = \frac{d}{dx} (xe^{-x}) = (1-x)e^{-x} \quad (1.2.10.10)$$

$$y'(\ln(2)) = (1 - \ln(2))e^{-\ln(2)} \quad (1.2.10.11)$$

$$= \frac{1 - \ln(2)}{2} \approx \frac{1 - 0.693147}{2} \approx 0.153 \quad (1.2.10.12)$$

b) Numerical Solution - Substituting equations (1.1.13.3) and (1.1.13.6) into our differential equation, we obtain:

$$\frac{y_{n+2} - 2y_{n+1} + y_n}{h^2} + 2\left(\frac{y_{n+1} - y_n}{h}\right) + y_n = 0 \quad (1.2.10.13)$$

Rearranging, we get:

$$y_{n+2} = 2(1-h)y_{n+1} - (1-h)^2y_n \quad (1.2.10.14)$$

Our initial conditions are:

$$y_0 = y(0) = 0, y_1 = y_0 + h(y'(0)) \quad (1.2.10.15)$$

We assume the following:

$$\ln 2 = 0.7, h = 0.01 \quad (1.2.10.16)$$

Then:

$$y_1 = 0 + 0.01(1) = 0.01 \quad (1.2.10.17)$$

Now, we continue this process until we find y_{70} and y_{71}

$$y_{70} = 0.3483, y_{71} = 0.3499 \quad (1.2.10.18)$$

From (1.1.13.3), we get:

$$y'_{70} = \frac{y_{71} - y_{70}}{h} \approx 0.145 \quad (1.2.10.19)$$

The following code simulates the difference between the numerical and theoretical solutions.

codes/Euler.py

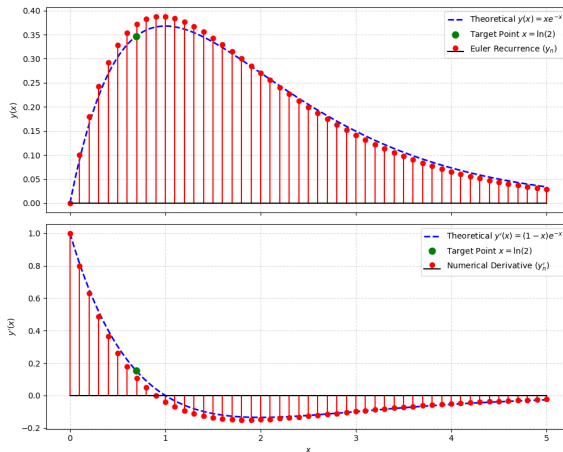


Fig. 1.2.10.1: Theoretical Plot vs Plot obtained with Euler's Forward Difference Method

1.2.11 Find the minimum value of the function $f(x) = |x| + |2x + 3|$ for real x . (AE 2026)

Solution: The critical points occur where the expressions inside the absolute values equal zero:

$$x = 0 \quad (1.2.11.1)$$

$$2x + 3 = 0 \implies x = -\frac{3}{2} = -1.5 \quad (1.2.11.2)$$

These two critical points divide the real line into three sub-intervals:

$$(-\infty, -1.5), \quad [-1.5, 0), \quad \text{and} \quad [0, \infty) \quad (1.2.11.3)$$

- For $x < -1.5$, both expressions inside the absolute values are negative:

$$f(x) = -x - (2x + 3) = -3x - 3 \quad (1.2.11.4)$$

- For $-1.5 \leq x < 0$, the term x is negative while $2x + 3$ is non-negative:

$$f(x) = -x + (2x + 3) = x + 3 \quad (1.2.11.5)$$

- For $x \geq 0$, both expressions are non-negative:

$$f(x) = x + (2x + 3) = 3x + 3 \quad (1.2.11.6)$$

Combining these cases gives the complete piecewise definition:

$$f(x) = \begin{cases} -3x - 3, & \text{if } x < -1.5 \\ x + 3, & \text{if } -1.5 \leq x < 0 \\ 3x + 3, & \text{if } x \geq 0 \end{cases} \quad (1.2.11.7)$$

Analyzing the behavior of each linear segment:

- On $(-\infty, -1.5)$, the line $f(x)$ has a negative slope (-3) , so $f(x)$ strictly decreases

as x approaches -1.5 .

- On $[-1.5, 0)$, the line $f(x)$ has a positive slope $(+1)$, so $f(x)$ strictly increases as x moves away from -1.5 .
- On $[0, \infty)$, the line $f(x)$ has a positive slope $(+3)$, so $f(x)$ continues to increase.

Since $f(x)$ decreases for $x < -1.5$ and increases for $x > -1.5$, the minimum value occurs at $x = -1.5$:

$$f(-1.5) = -1.5 + 3 = 1.5 \quad (1.2.11.8)$$

The following code plots the function.

codes/Modulus_function.py

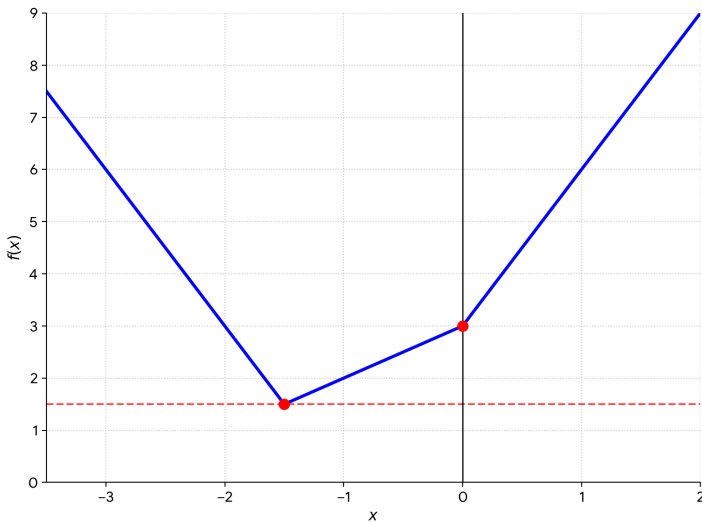


Fig. 1.2.11.1: Plot of the function

1.2.12 A bar, spring, and mass system is arranged in series.

- Bar: $E = 200$ GPa, $A = 100$ mm², $L = 100$ mm
- Spring Stiffness: $k = 200$ kN/mm
- Mass: $M = 100$ kg

Find the natural frequency of free vibration ω_n in rad/s. Also solve the differential equation obtained to obtain the position, velocity and acceleration of the system. (AE 2026)

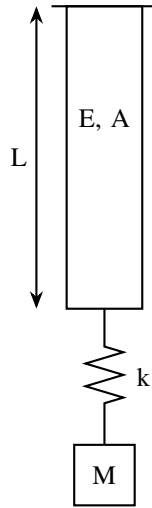


Fig. 1.2.12.1: Diagram of the system

Solution: The axial stiffness of the uniform bar is:

$$k_{\text{bar}} = \frac{AE}{L} \quad (1.2.12.1)$$

$$k_{\text{bar}} = \frac{10^{-4} \times 200 \times 10^9}{0.1} = 2 \times 10^8 \text{ N/m} \quad (1.2.12.2)$$

$$k = 200 \text{ kN/mm} = 2 \times 10^8 \text{ N/m} \quad (1.2.12.3)$$

Substituting the values of k and k_{eq} in (1.1.30.6) gives us

$$k_{\text{eq}} = \frac{k_{\text{bar}} \cdot k}{k_{\text{bar}} + k} = \frac{(2 \times 10^8) (2 \times 10^8)}{2 \times 10^8 + 2 \times 10^8} = 10^8 \text{ N/m} \quad (1.2.12.4)$$

The natural frequency of free vibration is:

$$\omega_n = \sqrt{\frac{k_{\text{eq}}}{M}} = \sqrt{\frac{10^8}{100}} = \sqrt{10^6} = 1000 \text{ rad/s} \quad (1.2.12.5)$$

Now, to find the position of the system as a function of time, from (1.1.30.14), we have:

$$\frac{d^2x}{dt^2} = \frac{-k}{M}x = -\omega_n^2 x \quad (1.2.12.6)$$

Let the initial displacement of the system be x_0 and initial velocity be v_0 . From result (1.1.12.6), we get:

$$x(t) = x_0 \cos(\omega_n t) + \frac{v_0}{\omega_n} \sin(\omega_n t) \quad (1.2.12.7)$$

Substituting the value of w_n ,

$$x(t) = x_0 \cos(1000t) + \frac{v_0}{1000} \sin(1000t) \quad (1.2.12.8)$$

Differentiating to get velocity:

$$v(t) = -1000x_0 \sin(1000t) + v_0 \cos(1000t) \quad (1.2.12.9)$$

Differentiating again to get the acceleration:

$$a(t) = -10^6 x_0 \cos(1000t) - 1000v_0 \sin(1000t) \quad (1.2.12.10)$$

The following code simulates the result.

codes/Spring.py

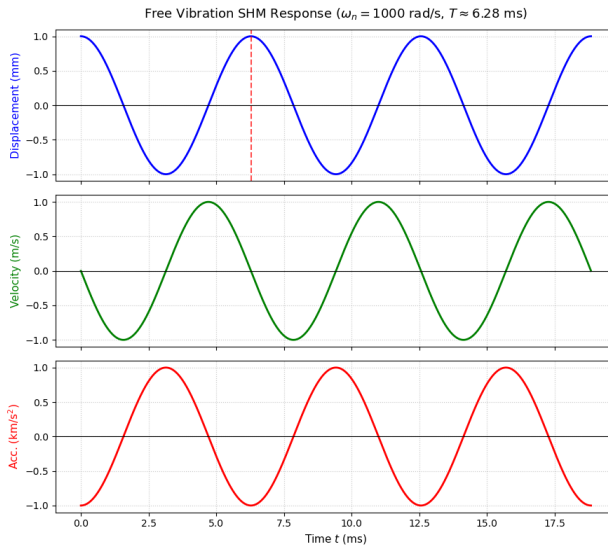


Fig. 1.2.12.2: Acceleration, Velocity and Position of the System